

## Polymer microarrays: Identification of substrates for phagocytosis assays

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### Abstract

A polymer microarray of 120 polyurethanes was used to identify polymers that promoted the adhesion of bone marrow dendritic cells (BMDC). Identified polymers were coated onto glass cover slips and shown to be efficient substrates for the immobilisation of these primary cells, which underwent efficient phagocytosis while still presumably maintaining their immature state.

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### 1. Introduction

Cellular adhesion has been investigated on a large number of substrates including glass, polystyrene, stainless steel, polypropylene and gold [1–3]. These materials are also often coated with various materials to promote cell binding, which include synthetic polymers (e.g. modified-polyethylenimine (PEI)) [2] and a variety of natural polymers ranging from poly L-lysine [3,4], fibronectin [2,5–7] and collagen [2,6] to more exotic materials such as spiders' silk [7]. Other approaches have included the use of monolayers of organic molecules coated onto gold substrates [8], but there are numerous (see for example [9–11]).

Cellular adhesion can be mediated by a variety of interactions, perhaps the most common being based on electrostatic interactions, with a highly positively charged surface leading to cellular immobilisation [2,4]. Such a surface is provided, at physiological pH, by coatings based on poly L-lysine and this has led to a myriad of applications and virtually universal use of poly L-lysine as a substrate for the immobilisation of cell lineages, but other poly-

cationic synthetic polymers such as PEI [12] have also been used to immobilise cells. Other materials commonly used, such as polystyrene, are believed to exert/mediate their effects by physiosorption of extracellular matrix proteins, which in turn promote interactions with membrane proteins and hence cell adhesion [2].

Dendritic cells (DC) play a central role in the initiation of immune responses and in the maintenance of tolerance to self [13]. As professional antigen presenting cells, they can engulf particulate matter, such as pathogens, necrotic and apoptotic cells by phagocytosis, process it and present it at the cell surface, bound to MHC class I or MHC class II molecules [14]. This ability means that DC are intensively studied as targets for vaccine design, particularly for vaccines against tumours. DC are a rare constituent of any organ; and one of the most common experimental sources is to purify the immature, highly phagocytic cells from mouse bone marrow dendritic cell (BMDC). However, immature murine BMDC are extremely sensitive to stimuli that cause maturation [15], which affects their ability to capture antigens by phagocytosis, while immobilisation is quite generally complicated by the fact that cellular behaviour may be modified by interactions with the materials used to coat the substrates [16,17].

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However, the immobilisation of DC would be important for several applications, ranging from simple phenotypic studies by microscopy, to the development of innovative cell-based assays. This paper shows how a polymer microarray [18] allowed the rapid and straightforward identification of specific polyurethanes able to act as substrates for cellular attachment of BMDC isolated from mouse. In addition we show that the identified polyurethane polymers compared more than favourably with the traditional adhesion compound, poly L-lysine in terms of cell adhesion and cellular behaviour.

## 2. Materials and methods

### 2.1. Synthesis of the polyurethane library [19]

Twelve parallel reactions were carried out in a Stem block on a mm scale by a typical two stage poly addition reaction. In brief, a pre-polymer was prepared by the reaction of 1.0 equivalent of a polyol with 2.0 equivalents of a diisocyanate in dry THF, followed (after titration) by the addition of 1.0 equivalent of a chain extender. The library was tailored by varying the nature of the polyols (250–2000 Da and different compositions); the diisocyanate and the chain extender and the ratio of chain extender/polyol. The monomers' structures used in the synthesis of the polymer library are shown in Fig. 1 together with an example of a representative polyurethane. The monomer used were: Poly(ethylene glycol) (PEG2000, PEG900, PEG400), poly(propylene glycol) (PPG2000,

PPG1000, PPG425), poly(tetramethylene glycol) (PTMG2000, PTMG1000, PTMG650, PTMG250), PEG-PPG-PEG (PEG-PPG1900), poly[1,6-hexanediol/neopentyl glycol/diethylene glycol-alt-(adipic acid)] diol (PHNAGD1800), [1,6-hexanediol/neopentyl glycol-alt-(adipic acid)] diol (PHNAD900) were purchased from Aldrich (Fig. 1) and dried in a vacuum oven at 50 °C for 24 h prior to use.

The 120 polymers of the library were all characterised using high-throughput methods such as GPC (column PLgel HTS-D 150 × 7.5 mm ID, Polymer Laboratories, 1-methyl-2-pyrrolidinone (NMP) 1 ml/min), Hyper DSC (Diamond, Perkin Elmer) and FT-IR (Mattson instrument). Polymers were named following a "PUnumber" format (see Supplementary Data for details).

### 2.2. Preparation of primary murine BMDC

C57BL/6 mice were bred locally and housed in standard facilities in Southampton. Female or male mice were used for bone marrow extraction. BMDC were purified from the femurs of 8–12 weeks old C57BL/6 mice, according to the method of Lutz et al. [20] and were used on day 10 of in vitro culture.

### 2.3. Polymer microarray preparation

Coating glass slides with agarose was achieved by dip-coating the aminoalkylsilane slide (Sigma-Aldrich) in a 1% w/v solution of agarose Type I-B (Sigma) at 65 °C followed by removal of the coating on the bottom side. After drying overnight at room temperature, the coated slides were stored at room temperature or used immediately for printing [18].

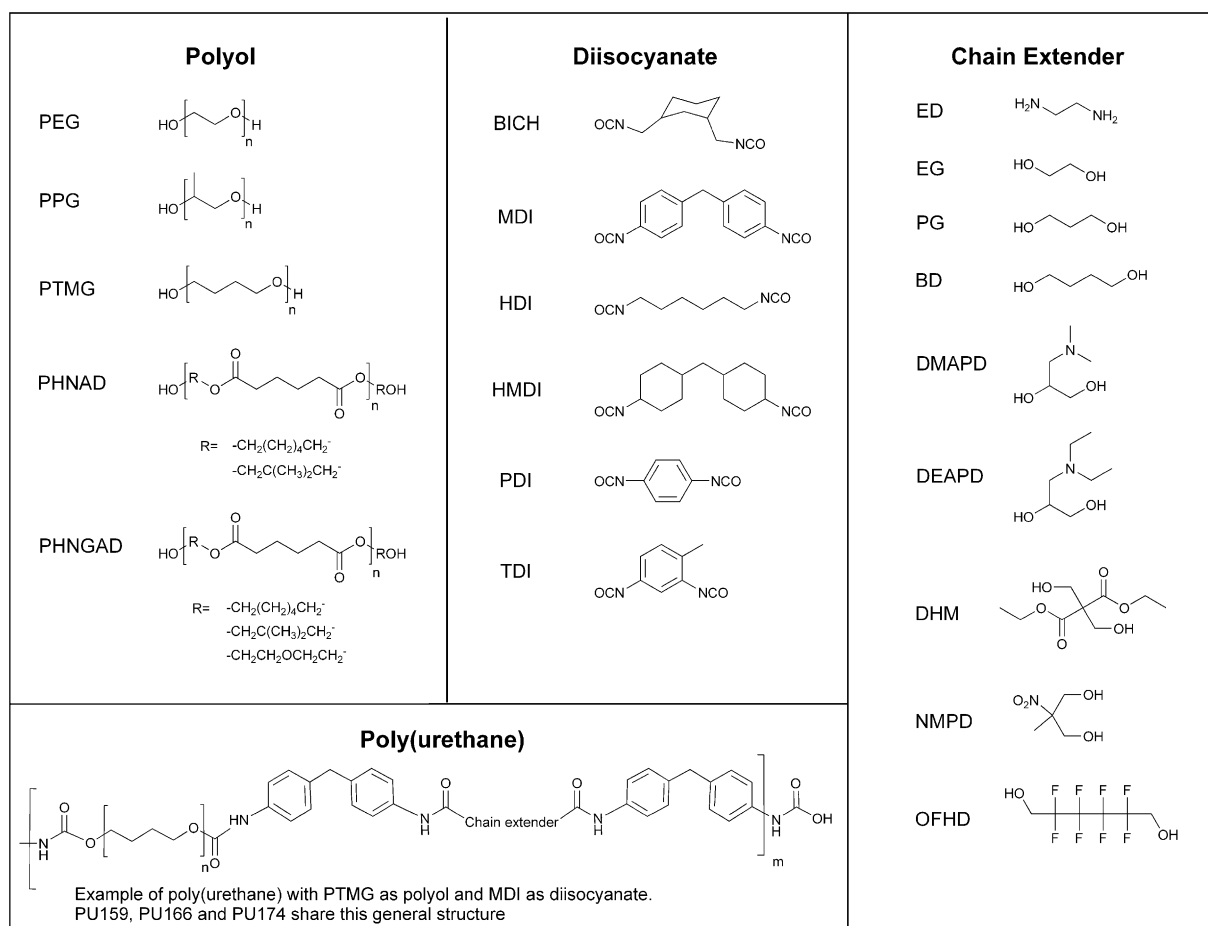


Fig. 1. Components used in the polyurethane library synthesis. A representative structure is shown. See Supplementary Data for details.

The polymer arrays were fabricated by contact printing (Genetix QArray mini; Hampshire, UK) with 16 aQu solid pins (K2785; Genetix; Hampshire, UK) using 1% w/v polymer solutions in NMP placed in polypropylene 384-wells microplates. The following printing conditions were used, 5 stamping per spot, 200 ms inking time and 10 ms stamping time. The typical spot size was 300–320  $\mu\text{m}$  in diameter with spot-to-spot distances of 1120  $\mu\text{m}$  allowing up to 480 polymers to be printed on a standard 25  $\times$  75 mm slide. Polymers were printed in quadruplicate within 2 fields of 16  $\times$  16 spots, while within each field a pattern of 4  $\times$  4 spots was left empty. Once printed, the slides were dried overnight under vacuum at 45 °C and were sterilised by exposure to UV irradiation for 20 min prior to use.

Prior to spin coating on a P6708 spin coater (Speedlines Technologies, IN, USA), 22 mm diameter glass cover slips were cleaned with tetrahydrofuran (THF). Fifty microlitres of the polymer solutions (2% w/v in THF) were placed onto the coverslips and spun for 10 s at 2000 rpm. Coverslips were dried under vacuum for 12 h and irradiated with UV light for 20 min before use.

#### 2.4. Cell-based microarray assays

BMDC were stained with 5  $\mu\text{M}$  carboxyfluorescein succinimidyl ester (CFSE, Molecular Probes, Invitrogen Detection Technologies, Renfrewshire, UK) to monitor cell adhesion as follows: an aliquot of 4  $\times$  10<sup>6</sup> cells was centrifuged for 5 min at 300g at 20 °C and washed once with phosphate-buffered saline (PBS) before resuspension in 1 ml PBS containing 5  $\mu\text{M}$  CFSE and incubating for 10 min at room temperature in the dark. After staining, the cells were centrifuged for 5 min at 300g at 20 °C and gently resuspended in 1 ml cell culture medium, R10: RPMI-1640 (Invitrogen, Renfrewshire, UK) supplemented with penicillin, streptomycin and L-glutamine (Invitrogen, 100 U/ml, 100  $\mu\text{g}/\text{ml}$ , 2 mM, respectively), 2-mercaptoethanol (50  $\mu\text{M}$ , Sigma) and 10% heat-inactivated and filtered low endotoxin foetal calf serum (Autogen Bioclear, Wiltshire, UK). The cells were gently pipetted onto the surface of a polymer microarray contained in sterile Petri dish. A further 20 ml R10 was carefully added to the dish, which was subsequently incubated at 37 °C with 5% CO<sub>2</sub> for 2 h. After gentle washing with R10 and then PBS, the cells were fixed with 4% w/v formaldehyde in PBS for 15 min at room temperature, then rinsed and stored in PBS at 4 °C. Adhesion was checked using a Zeiss Axiovert 200 fluorescence microscope. For precise quantification of cell adhesion, the fixed cells were further stained with a 0.5  $\mu\text{g}/\text{ml}$  solution of DAPI for 15 min at room temperature. Slides were rinsed and stored in PBS at 4 °C. Image capture and analyses were carried out using an IMSTAR high content screening (HCS) device equipped with the Pathfinder<sup>TM</sup> software (IMSTAR S.A., Paris, France). Cell compatibility with the different polymers was determined by automated counting of the number of cells present on each spot using both the DAPI and FITC channels.

#### 2.5. Analysis of BMDC

BMDC were resuspended in PBS containing 2% v/v HI-FCS and 5  $\mu\text{g}/\text{ml}$  phycoerythrin-conjugated hamster anti-CD11c monoclonal antibody (BD Pharmingen Oxfordshire, UK) or a phycoerythrin-conjugated isotype control antibody on ice for 30 min before washing three times with PBS/HI-FCS. Cells were immediately analysed by flow cytometry (FACSCalibur, BD, Oxfordshire, UK). For microscopy, BMDC adhered to coverslips were fixed in 4% v/v formaldehyde in PBS for 7 min at room temperature, permeabilised with 0.1% Triton X-100 for 7 min at room temperature, washed 3 times with PBS and then incubated with anti-CD11c monoclonal antibody for 1 h at room temperature. The coverslips were then washed 3 times with PBS before staining with a Quantum Dot 565 nm-conjugated anti-mouse secondary antibody (Quantum Dot, Cambridge, UK) and the nuclear dye, TO-PRO-3 (Molecular Probes, Invitrogen Detection Technologies, Renfrewshire, UK). Coverslips were mounted and analysed using a Leica SP2 laser scanning confocal microscope and a 100 $\times$  objective. Cells stained with secondary antibody

alone showed no detectable fluorescence using the same instrument settings.

Phagocytosis assays were carried out with coverslips coated with polymer PU159, PU166 or PU174 (as described above), or 0.01% w/v poly L-lysine solution (Sigma histology grade or Sigma tissue culture grade), or uncoated (incubated with PBS alone), were sterilised by UV light for 15 min in the bottom of a 6-well polystyrene culture plate (Greiner Bio-One, Gloucestershire, UK). A drop of 0.5 ml R10 containing 5  $\times$  10<sup>5</sup> BMDC was pipetted into the centre of each coverslip, and the plate incubated at 37 °C, 5% CO<sub>2</sub> for the cells to adhere, taking care not to disturb the meniscus. After 30 min, a further 1 ml of R10 was added to each coverslip, containing 5  $\mu\text{l}$  of 3.0  $\mu\text{m}$  diameter sulphate latex microspheres (IDC Latex, OR, USA), pre-coated with passively adsorbed foetal calf serum proteins. The microspheres were brought into contact with the adhered cells by centrifuging the plate for 3 min at 200g and 20 °C. The plate was then transferred to 37 °C, 5% CO<sub>2</sub> for a further 30 min.

After the incubation, the R10 was removed from the coverslips, which were rinsed briefly and gently with PBS, before being fixed with 4% formaldehyde in PBS for 7 min at room temperature. The BMDC were permeabilised by incubating with 0.1% v/v Triton X-100 in PBS for 7 min at room temperature, then washed 3 times with PBS before being stained with rabbit anti-calnexin antibody (Stressgen, CA, USA) and goat anti-rabbit AlexaFluor488 secondary antibody (Molecular Probes, Invitrogen Detection Technologies, Renfrewshire, UK). Nuclei were stained with TO-PRO-3.

Phagocytosis quantitation was carried out by examination of the coverslips using a Leica SP2 laser scanning confocal microscope with a 40 $\times$  objective. Three independent experiments were carried out, in which each coverslip was represented in duplicate. Phagocytosis was quantified by counting the number of microspheres internalised and the number of cells in a randomly selected field of view. Internalised microspheres were those defined as surrounded by calnexin staining in all three dimensions (Fig. 5). Figures for phagocytosis efficiency were derived from the counts of two to four fields of view per treatment, encompassing a minimum of 73 and a maximum of 556 cells. The degree of adhesion was classified by comparing the number of cells present in four randomly selected fields of view for each treatment and checking that these were typical over the three independent experiments.

### 3. Results and discussion

The aim of this project was to use a polymer microarray platform in order to discover new substrates for immobilising BMDC. The substrates so identified were used as per traditional coating systems on coverslips both to improve cellular adhesion and to allow different bioassays requiring cellular attachment to a 2D surface (such as phagocytosis and microscopy) to be carried out. The process is summarised in Fig. 2.

In this study 120 well-characterised polyurethanes were used. The advantages of this microarray platform are that it allows a common set of screening conditions for all polymers, requires only very limited amounts of each compound, and limited numbers of cells for each screen.

#### 3.1. Identification of materials for BMDC immobilisation using polymer microarrays

After extraction of BMDC from mice, cells were labelled with CFSE before incubating with the PU polymer microarray. Following fixation, slides were scanned and the number of cells per spot counted to determine which substrates showed good cellular compatibility with BMDC

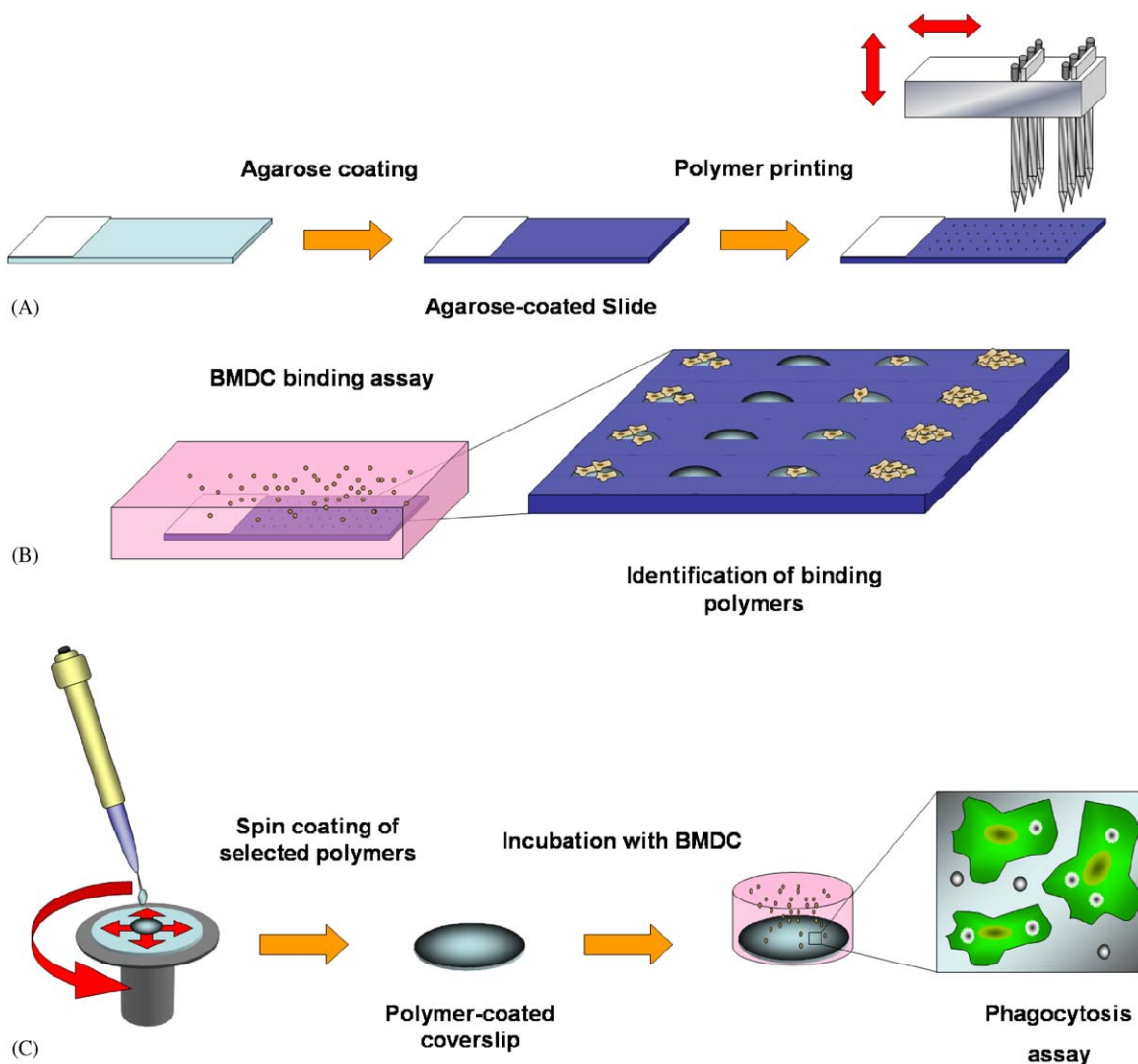


Fig. 2. General protocol for the identification of substrates for phagocytosis studies: (A) polymer microarray preparation; (B) microarray binding assay; (C) phagocytosis onto polymer coated coverslips.

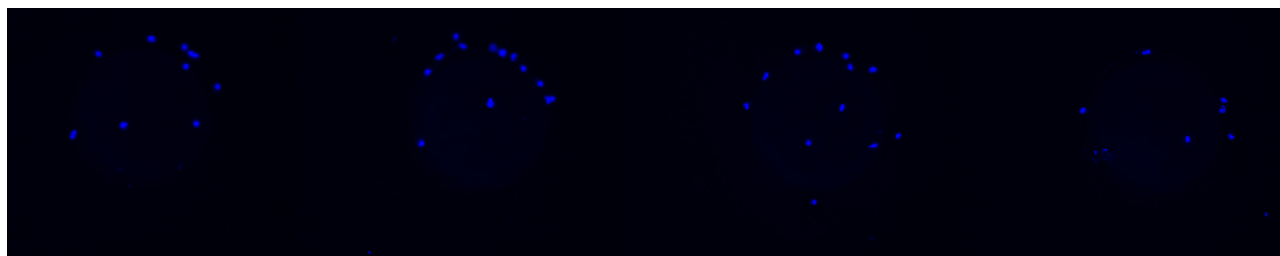


Fig. 3. Quadruplicate spots of a polymer showing good cellular adhesion. Images were taken with the HCS platform from IMSTAR with a  $20\times$  objective and DAPI channel.

(Fig. 3). The HCS platform from IMSTAR was used to scan the slides, allowing automated capture of single images for each single polymer spot with  $0.58\mu\text{m}$  resolution. This analysis also allowed polymers to be discarded which showed autofluorescence, avoiding the identification of false positives.

Three different polyurethanes (PU159, PU166 and PU174) immobilised more than 10 cells per spot (average

across 4 identical spots) and were selected and subsequently coated onto coverslips.

Prior to plating on coverslips, the purity of the BMDC preparation was assessed by staining cells with a fluorescent antibody against CD11c, a cell surface marker expressed by BMDC [20]. Stained cells were analysed by flow cytometry, which showed that greater than 90% of the cells were positive for CD11c (Fig. 4A). Cells also stained

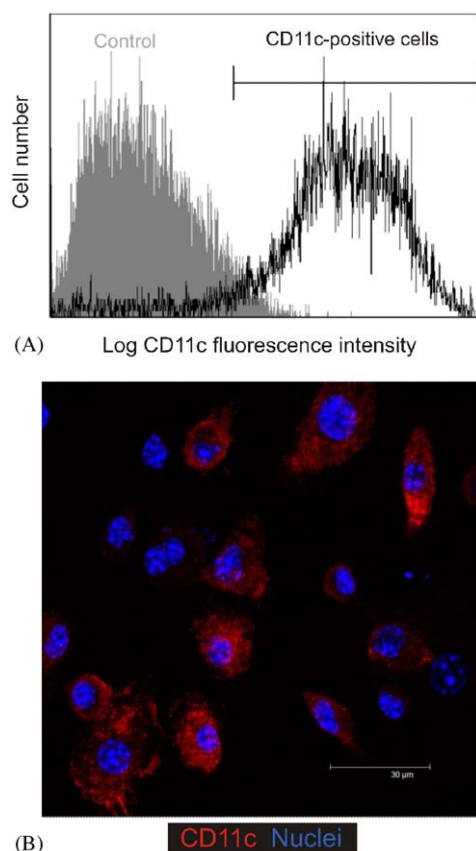


Fig. 4. (A) BMDC were stained with a fluorescently conjugated antibody against a dendritic cell surface marker, CD11c (black line), or with an isotype control antibody (shaded area, control), and analysed by flow cytometry. Cells staining positive for CD11c are indicated by a horizontal bar. (B) The anti-CD11c antibody (red) was used to stain BMDC that had adhered to the polymers (here, PU159), as described in Section 2. Nuclei (blue) were counterstained with TO-PRO-3. The image was obtained using a confocal laser scanning microscope and is a composite of 14, 0.5  $\mu$ m thick optical sections.

positive for the characteristic costimulatory markers CD80 and CD86 and were brightly stained by antibodies against major histocompatibility complex II (data not shown), all of which taken together indicate that the vast majority of cells in the preparation were BMDC. The CD11c antibody was then used to identify BMDC that had adhered to the polymer-coated coverslips. BMDC adhered rapidly and well to all three polymers, and the vast majority of adhered cells stained positive for CD11c (Fig. 4B). We cannot exclude that a very small number of the cells adhering to the polymers were contaminants, as BMDC preparations are never 100% pure.

### 3.2. Phagocytosis

Although BMDC could adhere well to the three polymers, it was possible that the cells were not able to phagocytose while immobilised. A comparison of phagocytic capacity was made between the three polymers and the traditional cell adhesion compound, poly L-lysine.

BMDC were plated and allowed to adhere to coverslips coated with either PU159, PU166, PU174, two grades of commercially available poly L-lysine or coverslips incubated in PBS alone. Cells were then supplied with 3  $\mu$ m diameter latex microspheres and incubated for 30 min at 37 °C. Post-incubation, the cells were fixed and stained for the endoplasmic reticulum protein, calnexin, which provides a convenient counterstain revealing the presence of internalised microspheres (Fig. 5). Phagocytic capacity was determined by confocal microscopy and counting of the number of microspheres that had been completely internalised (Table 1).

Interestingly, the greatest phagocytic capacity, an average of 4.4 microspheres per cell, was observed for cells that had adhered to coverslips treated with PBS alone (Fig. 5, Control panels). However, the degree of adhesion of cells was very poor, and the cells were liable to be washed away from the coverslip both during the assay and afterwards during staining. Poly L-lysine of both grades greatly improved adhesion, but there was a concomitant decrease in the phagocytic capacity to an average of 1.3 microspheres per cell (Fig. 5, Poly L-lysine panels). Although many microspheres were bound to the cell surface, closer examination of optical sections revealed that most had not been internalised. One possible explanation is that the physical restriction of cell movement caused by adhesion to a substrate inhibits the cytoskeletal rearrangements necessary during phagocytosis. In support of this idea, phagocytosis decreased to zero when the concentration of poly L-lysine was increased ten-fold (data not shown). Similar signalling pathways are involved for both cell adhesion to a substrate and adhesion to a particle to be phagocytosed [21]. Whereas cell adhesion can stimulate membrane extension, as for the filopodia and ruffles that engulf a particle, there are also cases where adhesion can inhibit membrane protrusion, such as the inhibition of cell migration in culture caused by cell–cell contacts [22], which may provide an alternative explanation for the observed decrease in phagocytosis by cells adhered to poly L-lysine. In addition to reducing the phagocytic capacity, poly L-lysine caused greater background binding of microspheres to the coverslip than was observed for the polymers and PBS control. This is most likely due to charge interactions, since the microspheres carry a net negative charge, and lysine is a positively charged amino acid at physiological pH.

By contrast, phagocytic capacity was much greater for the three polymers, with average values of 3.3 and 3.6 microspheres per cell for PU159 and PU174, respectively, although still less than that observed in the PBS control sample. Adhesion, measured as the number of cells in a field of view, was greater for PU166 than PU159 and PU174, but this was accompanied by a decrease in phagocytic capacity to 2.2 microspheres per cell. Taken together with the results for poly L-lysine, it appears that there is a trade-off between degree of adhesion and phagocytic capacity. Importantly however, adhesion to

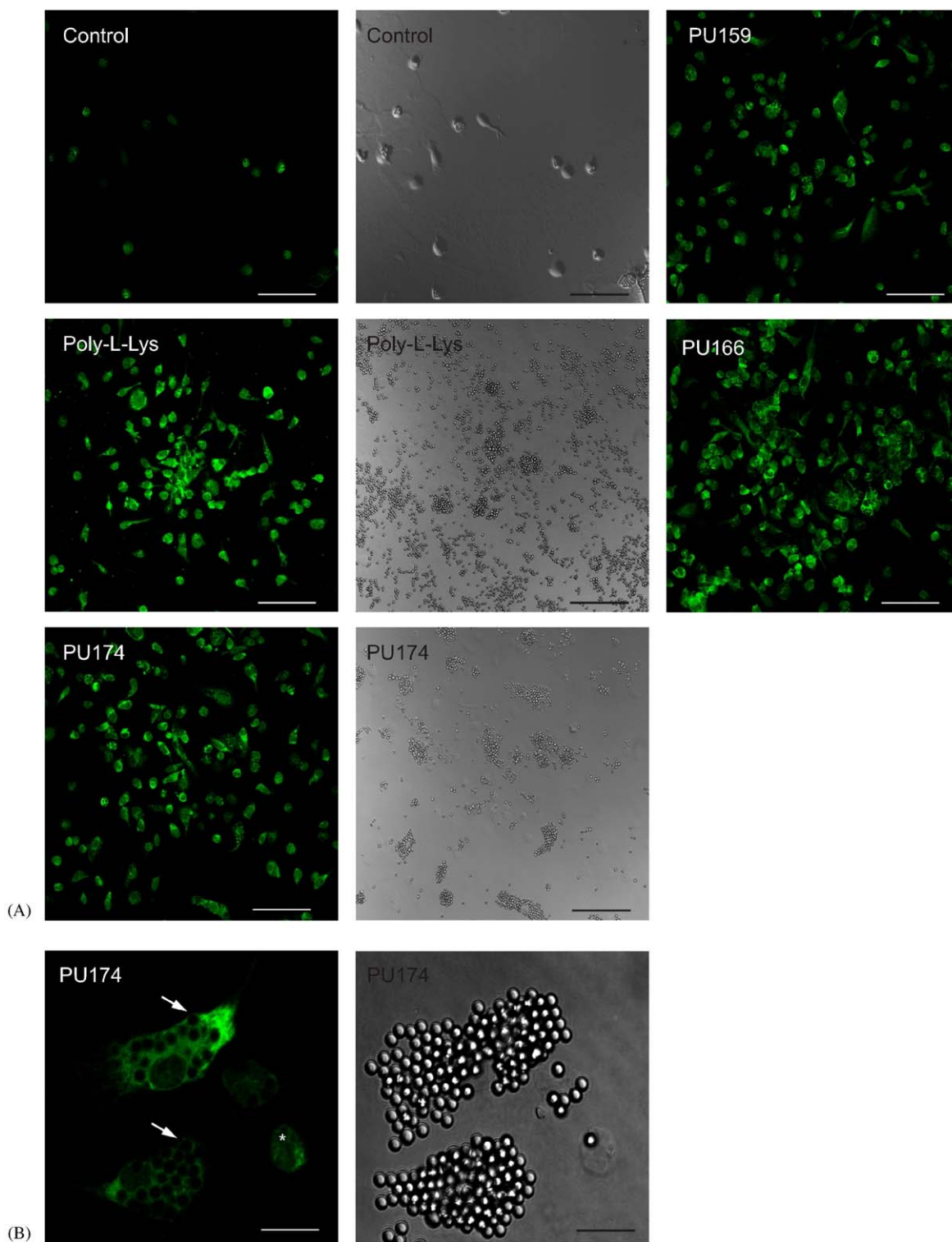


Fig. 5. Confocal laser scanning microscopy, with corresponding phase contrast images of BMDC adhered to coverslips and stained with anti-calnexin antibodies to reveal the presence of internalised microspheres. (A) Images taken with a 40 $\times$  objective. Control: coverslips treated with PBS alone; Poly L-Lys: coverslips treated with poly L-lysine; PU174, PU159 and PU166: coverslips coated with polyurethane polymers PU174, PU159 and PU166, respectively. Phase contrast images indicate the number of microspheres bound (but not necessarily internalised) to cells and also reveal the degree of non-specific binding of microspheres to the coverslip. Fluorescence images show optical sections 1.0  $\mu$ m thick. Horizontal scale bars represent 75  $\mu$ m. (B) Images taken with a 100 $\times$  objective. The fluorescence image shows in more detail the internalised microspheres (typical examples marked by arrows). The asterisk denotes the position of a microsphere that has bound to the cell surface (shown in the corresponding phase contrast image), but has not been internalised. Horizontal scale bars represent 15  $\mu$ m.

Table 1  
Phagocytosis results on different substrates

| Treatment                           | Adhesion  | Microspheres/cell | Cells counted | Microspheres counted |
|-------------------------------------|-----------|-------------------|---------------|----------------------|
| Control                             | Poor      | 4.4               | 73            | 324                  |
| Poly L-lysine, histology grade      | Good      | 1.3               | 312           | 421                  |
| Poly L-lysine, tissue culture grade | Excellent | 1.3               | 529           | 684                  |
| PU-159                              | Good      | 3.3               | 297           | 969                  |
| PU-166                              | Excellent | 2.2               | 556           | 1249                 |
| PU-174                              | Good      | 3.6               | 301           | 1073                 |

PU166 was as good as to tissue-culture grade poly L-lysine, but PU166 enabled almost double the phagocytic capacity. Our observations suggest that the PU polymers are superior to poly L-lysine, because they mediate good adhesion, while allowing much greater phagocytic activity.

#### 4. Conclusions

The high multiplexing power of the polymer microarray approach allows the screening of large numbers of different biomaterials under identical conditions in a single experiment, thus enabling the rapid determination of structure–activity relationships. Three polyurethanes were identified using the microarray approach which bound immature DC from murine bone marrows, all of which contained poly(tetramethylene glycol) (PTMG 250 or 1000 Da) as the diol and 4,4'-methylene bis(phenylisocyanate) (MDI) as the diisocyanate, with variation found in the chain extender (propyleneglycol, butane-1,4-diol or no chain extender).

This family of polymers will facilitate experimental handling of these cells and aid efforts in elucidating processes underlying antigen uptake, processing, presentation and immune stimulation or tolerisation and for studying intracellular processes such as cell migration and even interactions between cell types.

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#### Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biomaterials.2006.04.040.

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